Vivek Singh

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SUMMARY

Backend-focused Software Engineering candidate (B.E. Electronics and Telecommunication, 2027) with hands-on experience building Java and Spring Boot microservices, event-driven systems with Kafka, and low-latency services with Redis and MySQL. Solved 650+ Data Structures and Algorithms problems (LeetCode rating 1550+) and created Retrieval-Augmented Generation (RAG) applications with Python and LangChain.

EDUCATION

Army Institute of Technology, Pune, India — Aug 2023 – Jul 2027
Bachelor of Engineering in Electronics and Telecommunication — *CGPA: 8.17*

Army Public School, Gorakhpur, India — Jul 2021 – Jun 2022
Higher Secondary (CBSE) — *Percentage: 88.2%*

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, HTML, CSS
Frameworks: Spring Boot, Spring Framework, Spring Security, Spring Cloud, Hibernate/JPA
Backend & Architecture: REST APIs, Microservices, API Gateway, Service Discovery (Eureka), Event-Driven Architecture, Distributed Systems, WebSockets, Concurrency
Messaging & Caching: Apache Kafka, Redis
Databases: MySQL, MongoDB, ChromaDB (Vector Database)
AI / LLM: Retrieval-Augmented Generation (RAG), LangChain, OpenAI Whisper, Sentence Transformers, Streamlit
Core Concepts: Data Structures and Algorithms, Object-Oriented Programming (OOP), DBMS, System Design
Developer Tools: Git, GitHub, Docker, Docker Compose, Postman

PROJECTS

TradeX – Distributed Crypto Trading Engine | *Java, Spring Boot, Kafka, MySQL, Eureka, API Gateway* GitHub
- Architected 3 Spring Boot microservices (Order, Wallet, Trade) with Eureka discovery and 1 API Gateway entry point.
- Engineered a matching engine on 2 priority queues with O(log n) matching, price-time priority and partial fills.
- Decoupled all 3 services with Kafka events so trade settlement runs asynchronously.
- Designed a 2-balance wallet (available, locked) with locking to prevent double spending on concurrent orders.
- Developed REST APIs for 3 order lifecycle stages (placement, matching, execution) plus trade history.

AI Video Assistant (RAG) | *Python, Streamlit, LangChain, Whisper, ChromaDB, Mistral AI* GitHub
- Created a Retrieval-Augmented Generation (RAG) video assistant with a 6-stage pipeline for 2 input sources.
- Integrated OpenAI Whisper to transcribe 2 languages (English, Hindi), using yt-dlp and ffmpeg for audio chunking.
- Assembled a LangChain RAG chain with ChromaDB and Sentence Transformers generating 5 structured outputs and Q&A.

Ride-Sharing System (Uber-like Backend) | *Java, Spring Boot, Kafka, Redis, MySQL, WebSockets* GitHub
- Implemented nearest-driver discovery with Redis GEO indexing, achieving sub-50 ms lookups in project benchmarks.
- Processed 2 event streams (driver locations, trip events) asynchronously through Kafka.
- Prevented conflicting driver assignments with transactional state and locking, ensuring 1 active trip per driver.
- Persisted the full trip lifecycle (request, assignment, completion) across 3 stages in MySQL.
- Streamed live driver locations over 1 WebSocket connection per client, replacing client-side polling.

ACHIEVEMENTS

- Ranked 320 globally among 35,000+ participants in CodeChef Starters 225. Profile
- Solved 650+ Data Structures and Algorithms problems across LeetCode and CodeChef.
- Achieved a LeetCode contest rating of 1550+ through regular participation in weekly contests. Profile

LEADERSHIP EXPERIENCE

Joint Secretary, Coding Club — Jun 2024 – Present
Army Institute of Technology, Pune
- Organized 5+ coding contests and DSA sessions for 150+ students, promoting competitive programming.
- Led a 10-member core team in planning and executing coding events and DSA initiatives.
- Engaged 120+ students through competitive programming and regular LeetCode and Codeforces practice.